# InsightStream

## Data Flywheel & Moat Plan

*Buy vs. Bake Analysis • 3-Step Flywheel • Proprietary Data Moat • Unit Economics*

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# 1. Buy vs. Bake Analysis

The first strategic decision in any AI product is whether to use a third-party model (Buy) or train your own (Bake). This is not a permanent choice. It is a phase-dependent decision that should be re-evaluated as the product scales.

## 1.1 Why Buy for v1: Gemini 2.5 Flash

| Factor | Buy (Gemini 2.5 Flash API) | Bake (Fine-tuned Llama 3 8B) |
|---|---|---|
| **Time to market** | **Weeks. API key + prompt = working pipeline.** | **Months. Requires labeled dataset, training infra, GPU allocation, evaluation.** |
| **Upfront cost** | **$0. Pay-per-use only.** | **$5K–$50K+ for GPU hours (A100/H100), depending on dataset size.** |
| **Per-inference cost** | **~$0.02 per 60-min interview.** | ~$0.005–$0.01 self-hosted. Cheaper at scale, but requires GPU infrastructure. |
| **Quality (zero-shot)** | **High. Gemini 2.5 Flash handles structured output, multilingual, sarcasm out of the box.** | **Low. Llama 3 8B without fine-tuning produces inferior structured extraction vs. Gemini.** |
| **Quality (fine-tuned)** | N/A — no fine-tuning available for Flash. | **Potentially superior for InsightStream-specific patterns after 10K+ training pairs.** |
| **Context window** | **1M tokens. Handles 3-hour interviews.** | **8K–32K tokens. Requires chunking for anything over ~20 min.** |
| **Data privacy** | **Data sent to Google API. Governed by Google's ToS.** | **Full control. No data leaves your infrastructure.** |
| **Labeled data required** | **None. Prompt engineering only.** | **Minimum 5K–10K labeled pairs for meaningful improvement.** |
| **Vendor lock-in risk** | **High. Dependent on Google's pricing, availability, and model behavior.** | **None. You own the model weights.** |

## 1.2 When to Bake: Trigger Conditions

The decision to fine-tune a proprietary model should be triggered by concrete, measurable thresholds not aspiration. InsightStream should switch from Buy to Bake when ALL three conditions are met simultaneously:

| # | Trigger Condition | Threshold | Why This Number |
|---|---|---|---|
| **1** | Correction pair volume | **≥ 10,000 labeled pairs** | Below this, fine-tuning produces minimal improvement over prompt engineering. Industry standard for domain adaptation. |
| **2** | Monthly API cost exceeds self-hosting | **> $2,000/month sustained** | At $0.02/interview, this means ~100K interviews/month. Below this volume, API is cheaper than GPU infra. |
| **3** | Accuracy plateau on Gemini | **Manual Override Rate stuck > 20% for 60+ days** | If prompt engineering cannot push override rate below 20%, the model needs domain-specific training — not better prompts. |

**Current assessment:** InsightStream is at 0 correction pairs, <$5/month API cost, and pre-launch override rate is unmeasured. All three triggers are far from being met. Buy is the correct strategy for the next 6–12 months.

## 1.3 Bake Migration Path

When trigger conditions are met, the migration plan is:

1. Export the anonymized correction pair dataset from the telemetry database (opt-in users only).
2. Fine-tune Llama 3 8B (or current best open-weight model) using the correction pairs as (instruction, preferred_output, rejected_output) DPO/RLHF triplets.
3. Run the full evaluation harness (Golden Dataset + hard-mode pairs) on the fine-tuned model.
4. A/B test: route 10% of traffic to the fine-tuned model, 90% to Gemini. Compare Manual Override Rate, Precision @ K, and latency.
5. If the fine-tuned model matches or beats Gemini on all three metrics, promote it to 100% and deprecate the Gemini dependency.
6. Maintain Gemini as a fallback for users who prefer it (via model selector dropdown).

# 2. The 3-Step Data Flywheel

InsightStream's competitive advantage is not the pipeline architecture (which can be replicated) or the base model (which is a public API). It is the proprietary dataset generated by human corrections that no competitor can access. This dataset compounds with every user interaction, creating a self-reinforcing feedback loop.

## 2.1 Flywheel Diagram

**STEP 1: USAGE — PM analyzes interview transcripts**

▼ generates correction signals

**STEP 2: LABELING — Implicit + Explicit feedback captured**

▼ correction pairs accumulated

**STEP 3: RE-INVESTMENT — Data improves model accuracy**

▼ better accuracy → lower override rate → more usage

**↻ FLYWHEEL REPEATS — Each cycle compounds the data moat**

## 2.2 Step 1: Usage (Data Generation)

Every time a PM uploads a transcript and clicks "Analyze," the pipeline generates structured AI outputs (feature requests, pain points, themes). These outputs are hypotheses, not facts. The value is created in the next step, when the human corrects them.

**Data generated per run:** N feature requests + M pain points + K themes, each with a verbatim quote, confidence score, and source_id attribution. A typical 30-minute interview produces 5–15 insights.

**Volume projection:** A single PM analyzing 5 interviews/week generates ~40–75 AI-produced insights per week. A team of 10 PMs generates 400–750 per week. At this rate, the 10,000-pair fine-tuning threshold is reached in approximately 3–6 months.

## 2.3 Step 2: Labeling (Feedback Capture)

InsightStream captures two categories of correction signals, both without requiring any extra effort from the user:

### Implicit Feedback (Behavioral Signals)

| User Action | Signal | Training Value |
|---|---|---|
| Exports insight to CSV (checkbox checked) | APPROVE — user agrees with AI extraction | Positive training pair: (transcript_segment, insight) → GOOD |
| Unchecks "Approve for Backlog" before export | REJECT — user disagrees with AI extraction | Negative training pair: (transcript_segment, insight) → BAD |
| Uses "Missing Something?" search bar | COVERAGE GAP — pipeline missed something | Implicit negative: the search query text reveals what the user expected but didn't get |
| Clicks "Swap Speakers" button | DIARIZATION ERROR — labels were inverted | Preprocessing signal: correlate with audio characteristics for future auto-correction |
| Time spent reviewing each insight (future) | DIFFICULTY SIGNAL — longer review = harder case | Weight harder cases higher in fine-tuning loss function |

### Explicit Feedback (Direct Corrections)

| User Action | Signal | Training Value |
|---|---|---|
| Edits an insight before export (future feature) | CORRECTION — user rewrites the AI's output | Highest-value pair: (original_AI_output, human_corrected_output) → DPO preferred/rejected |
| Rates insight quality (future: thumbs up/down) | QUALITY RATING — binary signal | Reinforcement signal for RLHF or reward model training |
| Adds a manual insight not extracted by AI (future) | MISSED INSIGHT — ground truth the AI should have found | Direct supervision signal for recall improvement |

**Key design principle:** The highest-value signals come from normal workflow actions (approve/reject/search), not special feedback UI. Users never feel like they're "training the AI." They're just doing their job. The training data is a byproduct.

## 2.4 Step 3: Re-Investment (Model Improvement)

Correction pairs are re-invested into the system through three channels, ordered by implementation complexity:

### Channel 1: Dynamic Few-Shot Injection (v1 — Immediate)

**Mechanism:** Select the 3–5 most recent high-confidence correction pairs relevant to the current transcript's domain and inject them as few-shot examples into the Extractor prompt.

**Implementation:** Before each extraction run, query the correction log for pairs where correction_type = APPROVE and the transcript domain matches (based on keyword overlap). Prepend these as "Here are examples of well-extracted insights from similar interviews" in the prompt.

**Expected impact:** 5–15% improvement in extraction accuracy for domain-specific patterns (e.g., SaaS onboarding interviews, developer tool feedback). No model training required.

### Channel 2: Prompt Optimization via A/B Testing (v2 — 3 months)

**Mechanism:** Use the accumulated correction pairs to systematically test prompt variants. Measure Manual Override Rate for each variant against the same transcript set.

**Implementation:** Run 2–3 prompt variants per week on incoming transcripts. Track which variant produces the lowest override rate. Promote winners to default.

**Expected impact:** 10–20% reduction in Manual Override Rate over 3 months of continuous optimization.

### Channel 3: Model Fine-Tuning (v3 — 6–12 months)

**Mechanism:** Export anonymized correction pairs as DPO (Direct Preference Optimization) triplets: (instruction, preferred_output, rejected_output). Fine-tune Llama 3 8B or equivalent open-weight model.

**Implementation:** Requires the secure telemetry opt-in stream to accumulate 10K+ pairs. Fine-tuning job runs on cloud GPU infrastructure. A/B tested against Gemini before promotion.

**Expected impact:** 20–40% reduction in per-inference cost (self-hosted vs. API) plus domain-specific accuracy improvements that prompt engineering alone cannot achieve.

# 3. The Proprietary Data Moat

A competitor can replicate InsightStream's architecture in weeks. They can use the same Gemini API, the same LangGraph framework, and the same Pydantic schemas. What they cannot replicate is the correction dataset.

## 3.1 What InsightStream Owns That Competitors Do Not

| Data Asset | Description | Why a Competitor Cannot Replicate |
|---|---|---|
| **Human-Corrected Insight Pairs** | (transcript_segment, AI_insight, human_decision) triplets from the Verification Center | Requires the same user base making the same corrections. Cannot be purchased, scraped, or synthesized. |
| **Coverage Gap Queries** | The exact text of every "Missing Something?" search query | Reveals what users expected but the pipeline missed. This is a negative-space dataset that maps blind spots. |
| **Diarization Error Patterns** | Correlation between audio characteristics and speaker swap corrections | Requires users to actually click "Swap Speakers" on real interviews. No synthetic substitute. |
| **Confidence Calibration Data** | Mapping of AI confidence scores to actual human approval rates | "Score 7 gets approved 92% of the time; score 4 gets approved 61%." This calibration curve is unique to our user base. |
| **Domain-Specific Extraction Patterns** | Which insight structures (user_need + proposed_solution combinations) are approved in SaaS PM contexts | Encodes the implicit standards of professional product management. A general-purpose LLM doesn't know these. |

## 3.2 Moat Depth Over Time

| Timeframe | Correction Pairs | Moat Depth | Competitive Advantage |
|---|---|---|---|
| Month 0–3 | 0 – 1,000 | Shallow. Architecture is the differentiator. | First-mover advantage only. A competitor can catch up in weeks. |
| Month 3–6 | 1,000 – 5,000 | Moderate. Few-shot injection shows measurable improvement. | Competitor would need 3–6 months of user activity to match. Meaningful head start. |
| Month 6–12 | 5,000 – 10,000 | Deep. Fine-tuning threshold reached. | Competitor needs both the user base AND the correction volume. 6–12 month gap. |
| Month 12+ | 10,000+ | Structural. Proprietary model outperforms generic API. | Competitor cannot close the gap without equivalent user base. The data compounds. |

# 4. Unit Economics & ROI

## 4.1 Inference Cost per Interview

| API Call | Input Tokens | Output Tokens | Cost |
|---|---|---|---|
| Global Context Summary | ~4,000 | ~100 | $0.0006 |
| Extractor (per chunk, assume 2 chunks) | ~3,000 × 2 | ~500 × 2 | $0.0015 |
| Deduplicator | ~2,000 | ~1,000 | $0.0009 |
| Auditor (per insight, assume 8 insights) | ~800 × 8 | ~10 × 8 | $0.0010 |
| Synthesizer | ~2,000 | ~500 | $0.0006 |
| **TOTAL PER 60-MIN INTERVIEW** | **~18,400** | **~2,180** | **~$0.02** |

## 4.2 Value Realized

| Metric | Manual Process | InsightStream |
|---|---|---|
| Time per interview analysis | **4 hours** | **45 seconds + 10 min review** |
| PM hourly rate (loaded cost) | $75–$150/hour | $75–$150/hour |
| Cost per interview (labor) | **$300–$600** | **$12.50–$25 (review time only)** |
| API cost per interview | $0 | $0.02 |
| Total cost per interview | **$300–$600** | **$12.52–$25.02** |
| **ROI per interview** | **Baseline** | **24x–48x cost reduction** |

**Bottom line:** InsightStream costs $0.02 in API fees and ~15 minutes of PM review time to produce what previously required 4 hours of manual work. The ROI is not incremental. It is a category change in how research synthesis gets done.

## 4.3 Flywheel Efficiency: Accuracy Improvement Per 100 Interviews

| Interviews Processed | Correction Pairs | Improvement Channel | Expected Override Rate |
|---|---|---|---|
| 0–100 | ~500–1,000 | Baseline prompts only. No data re-investment yet. | ~20–25% (estimated pre-launch) |
| 100–500 | 1,000–3,000 | Few-shot injection begins. Top correction pairs injected into Extractor prompt. | ~15–20% (target: below 15%) |
| 500–2,000 | 3,000–10,000 | Prompt A/B testing at scale. Statistically significant variant selection. | ~10–15% |
| 2,000+ | 10,000+ | Fine-tuning triggered. Domain-specific model replaces generic API. | ~5–10% (target steady state) |

Each 100-interview increment produces approximately 500–1,000 correction pairs. The first 100 interviews establish the baseline and identify the most common extraction errors. Interviews 100–500 enable the first data-driven improvements. Beyond 500, the accuracy gains per additional interview begin to plateau, but the moat continues to deepen because a competitor starting from zero still needs to process those same 500 interviews to reach parity.

# 5. Cold Start Strategy

The flywheel creates compounding value over time. But the first user has zero historical data. InsightStream must deliver immediate value without any correction history. Here is how:

## 5.1 Day-Zero Value Proposition

InsightStream's v1 does not depend on the flywheel to be useful. The base pipeline (Gemini + Pydantic + Auditor) delivers value from the first use:

- The Extractor uses a carefully engineered system prompt with an embedded confidence rubric. No historical data is needed. It works on any English-language interview transcript from day one.
- The Auditor's semantic entailment check is a zero-shot capability of the base model. It does not require training data to enforce the 0% hallucination guarantee.
- The Verification Center provides human review controls regardless of model accuracy. Even if the AI misses 50% of insights, the user still saves time because the other 50% is pre-extracted with citations.
- The "Missing Something?" RAG search bar catches anything the pipeline misses. The user always has a fallback.

## 5.2 Bootstrapping the Flywheel

The cold start problem is not "how do we provide value?" (the base pipeline handles that). It is "how do we start generating correction pairs as fast as possible?"

1. Seed with internal usage: Before public launch, run InsightStream on 20–50 internal interviews. Have team members review the outputs in the Verification Center. This generates 100–500 initial correction pairs before any external user touches the system.
2. Prioritize high-correction-rate users: Users who actively reject and search (high override rate) generate the most valuable training data. Identify these users and ensure their correction data is captured with high fidelity.
3. Publish the LinkedIn portfolio launch: The initial announcement targets PMs who will use the tool on real interviews, generating real corrections. Even 10 active users processing 5 interviews/week generate 250–500 correction pairs per week.
4. Offer the tool free during the data-accumulation phase: The API cost is ~$0.02/interview. Subsidizing 10,000 interviews costs $200. The correction dataset generated is worth orders of magnitude more than the API spend.

## 5.3 Cold Start to Flywheel Transition Timeline

| Week | Activity | Correction Pairs | Flywheel Status |
|---|---|---|---|
| Week 0–2 | Internal dogfooding (20–50 interviews) | 100–500 | Cold start. Base prompt only. |
| Week 2–4 | LinkedIn launch. First 10 external users. | 500–1,500 | Data accumulating. No re-investment yet. |
| Week 4–8 | Few-shot injection enabled. Top pairs added to prompts. | 1,500–3,000 | **Flywheel turning. First accuracy gains visible.** |
| Week 8–16 | Prompt A/B testing begins. | 3,000–7,000 | **Flywheel accelerating. Override rate dropping.** |
| Week 16–24 | Fine-tuning evaluation (if 10K pairs reached). | 7,000–15,000 | **Flywheel self-sustaining. Moat established.** |

**The key insight:** The cold start is not a problem to solve. It is a phase to move through as quickly as possible. The base product is valuable from day one. The flywheel makes it more valuable every day after that. The goal is to reach 10,000 correction pairs before a competitor reaches 1,000.